OPENSOURCESDRLAB DOCUMENTATION

ZYNQ7020-FMC Core Module Schematic

Product	ZYNQ7020-FMC
Document	OSSDRLAB-ZYNQ7020-FMC-SCH-CORE
Version	1.0
Date	2026-11-12

This document contains the circuit schematic of the ZYNQ7020 core module used on the OpenSourceSDRLab ZYNQ7020-FMC development board, exported from Altium Designer (A4). The eight schematic sheets follow this cover page.

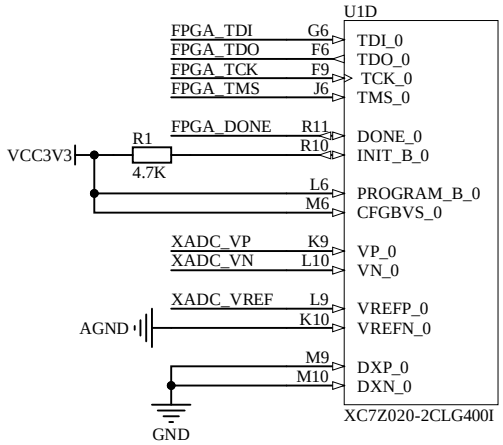
Sheet	Content
1	Zynq configuration bank 0: JTAG, DONE / INIT_B / PROGRAM_B, XADC with REF3312 voltage reference
2	PS MIO banks 500/501, 33.333 MHz PS clock, PS reset, W25Q256JV quad-SPI flash (32 MB), boot-mode straps
3	PL I/O banks 13, 34 and 35
4	PS DDR3 interface (bank 502) and TPS51200 DDR termination regulator
5	DDR3 memory: two MT41K256M16TW (2 × 4 Gbit, 32-bit bus, 1 GB) and termination resistors
6	Zynq power pins and decoupling
7	Board-to-board connectors to the carrier (2 × 120-pin)
8	Power supply: four MP2145 buck regulators, power LED

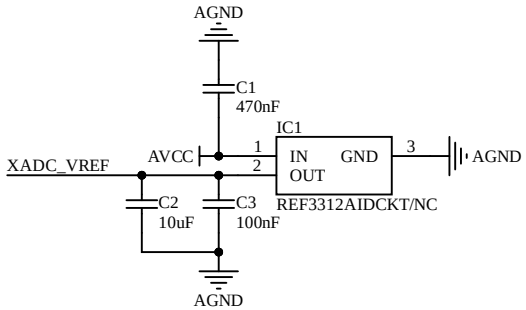
Device: Xilinx/AMD Zynq-7000 XC7Z020-2CLG400I.

Repository: https://github.com/OpenSourceSDRLab/ZYNQ7020-FMC

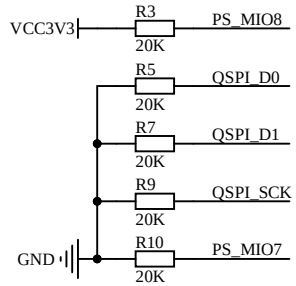
Questions about the hardware: support@opensourcesdrlab.com

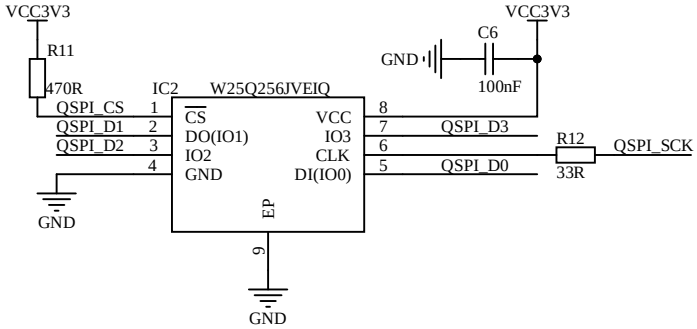
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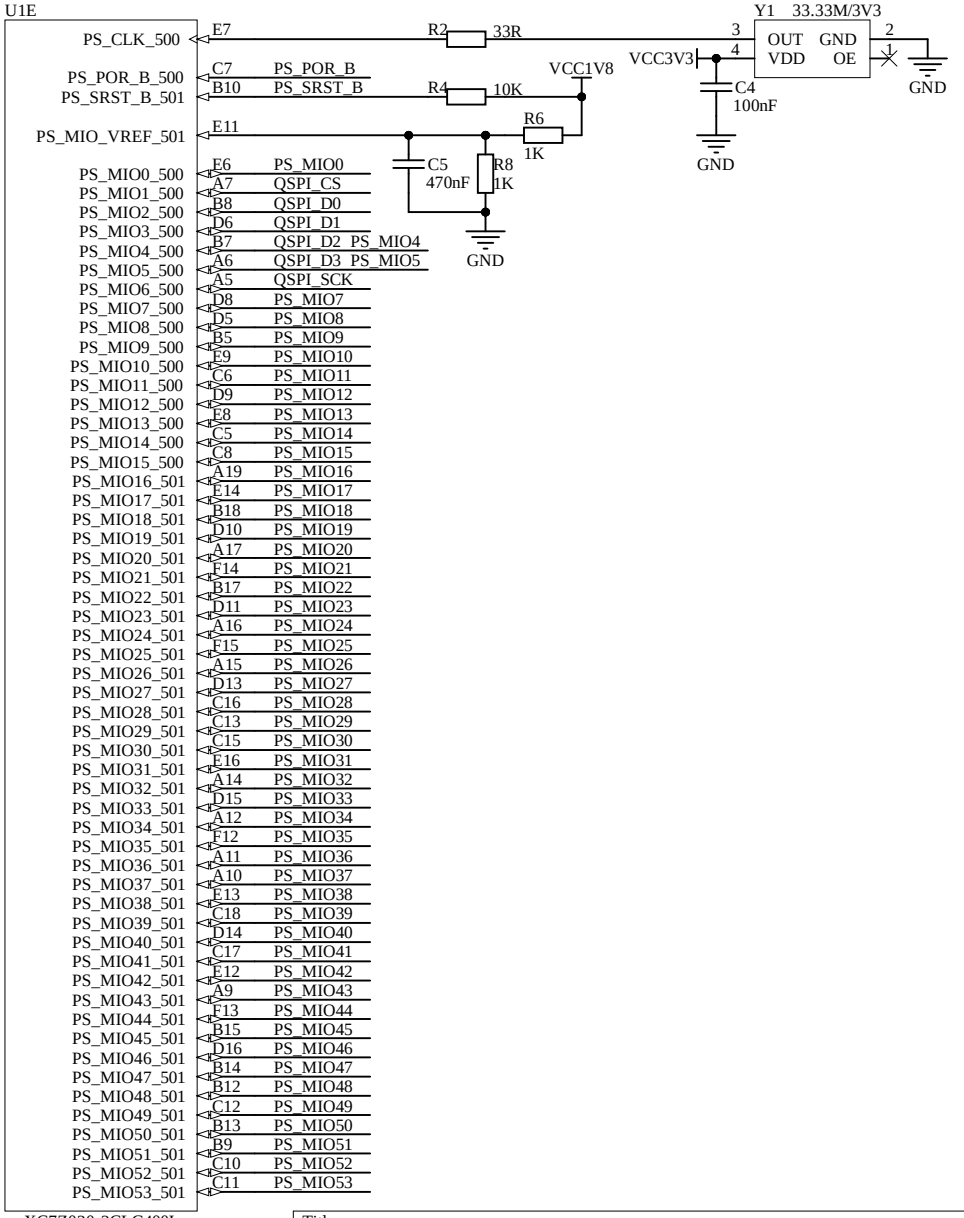




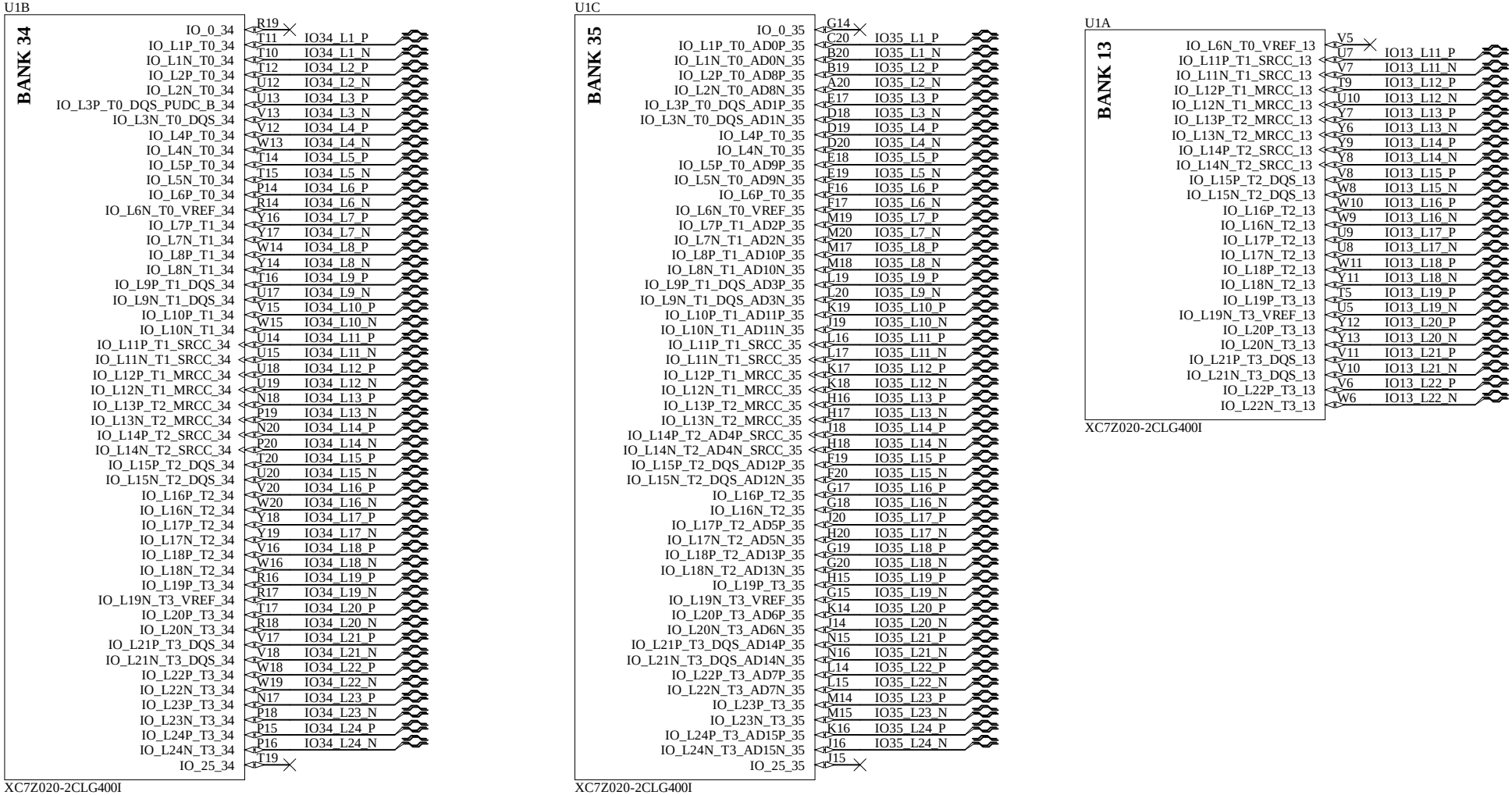
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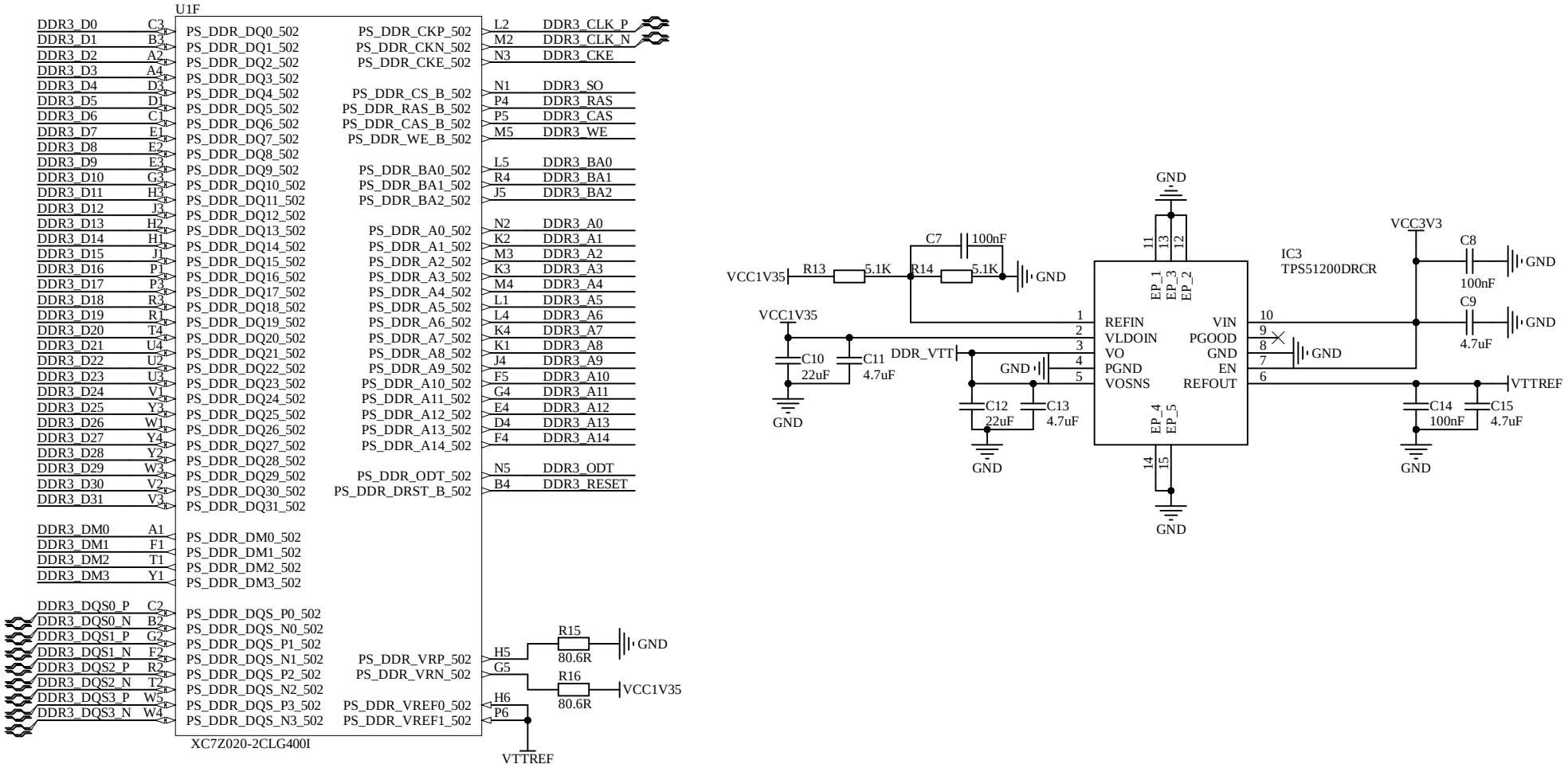




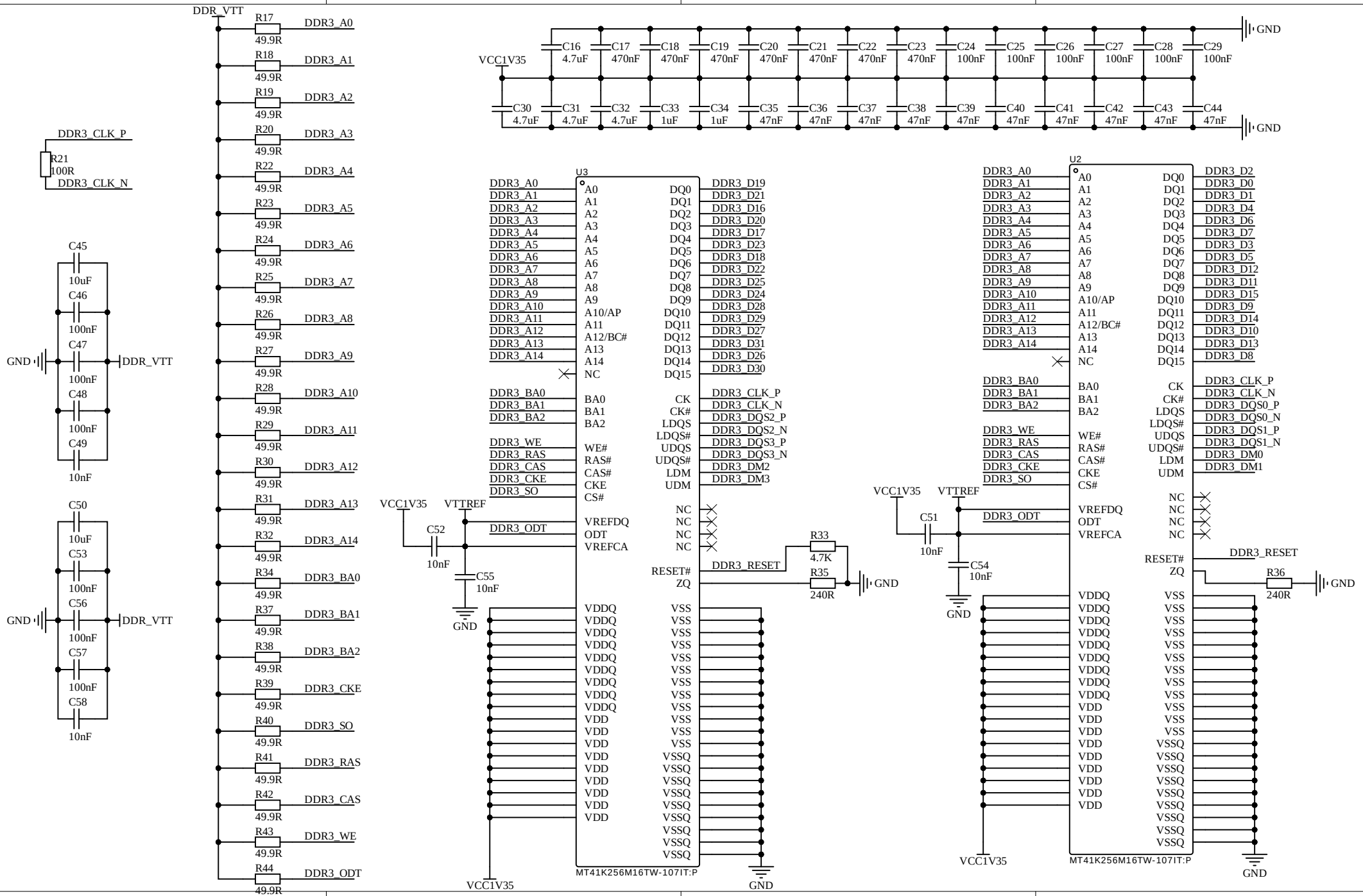


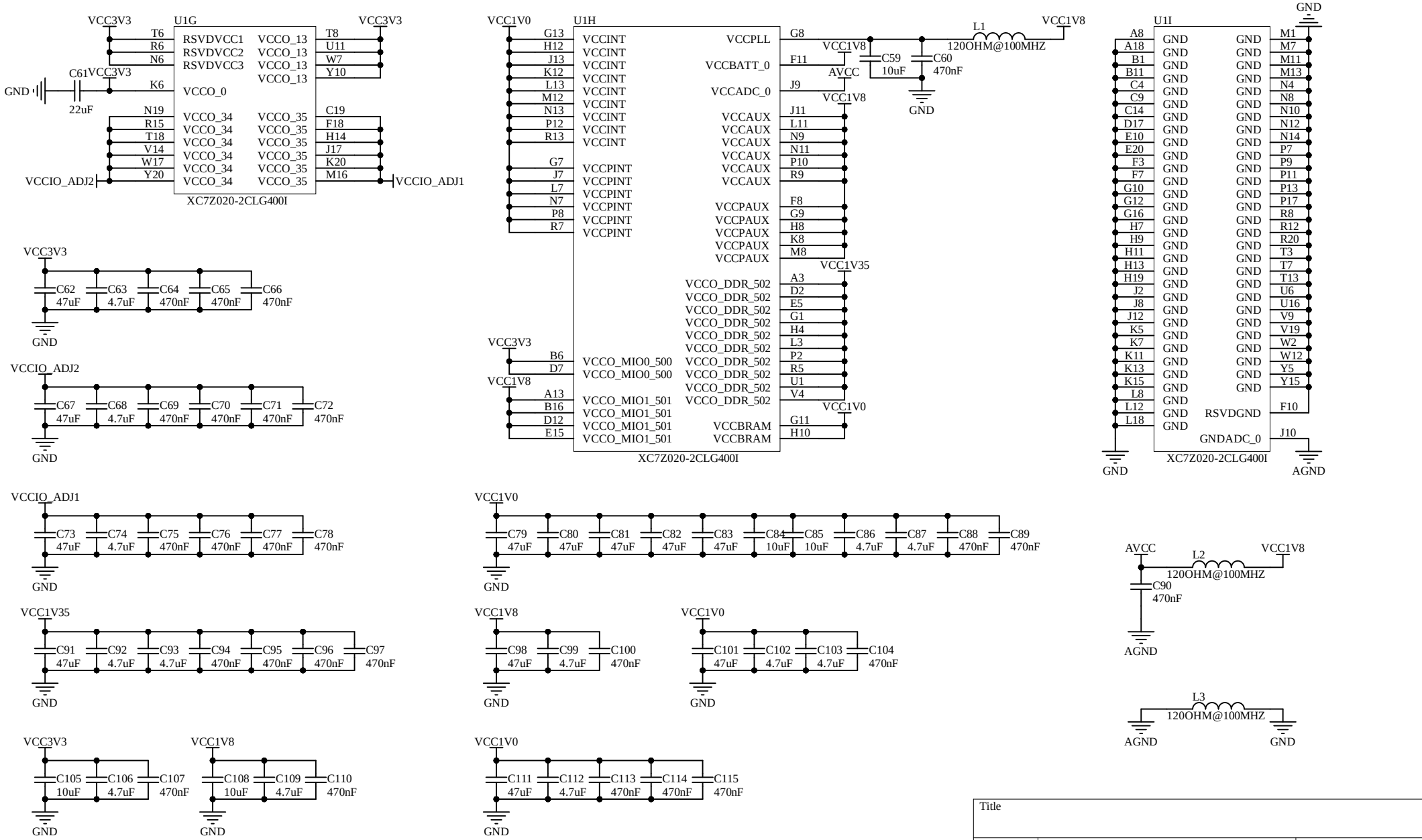
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U1B
BANK 34
IO_0_34 R19
IO_L1P_T0_34 T11 IO34_L1_P
IO_L1N_T0_34 T10 IO34_L1_N
IO_L2P_T0_34 T12 IO34_L2_P
IO_L2N_T0_34 U12 IO34_L2_N
IO_L3P_T0_DQS_PUDC_B_34 U13 IO34_L3_P
IO_L3N_T0_DQS_34 V13 IO34_L3_N
IO_L4P_T0_34 V12 IO34_L4_P
IO_L4N_T0_34 W13 IO34_L4_N
IO_L5P_T0_34 T14 IO34_L5_P
IO_L5N_T0_34 T15 IO34_L5_N
IO_L6P_T0_34 P14 IO34_L6_P
IO_L6N_T0_VREF_34 R14 IO34_L6_N
IO_L7P_T1_34 Y16 IO34_L7_P
IO_L7N_T1_34 Y17 IO34_L7_N
IO_L8P_T1_34 W14 IO34_L8_P
IO_L8N_T1_34 Y14 IO34_L8_N
IO_L9P_T1_DQS_34 T16 IO34_L9_P
IO_L9N_T1_DQS_34 U17 IO34_L9_N
IO_L10P_T1_34 V15 IO34_L10_P
IO_L10N_T1_34 W15 IO34_L10_N
IO_L11P_T1_SRCC_34 U14 IO34_L11_P
IO_L11N_T1_SRCC_34 U15 IO34_L11_N
IO_L12P_T1_MRCC_34 U18 IO34_L12_P
IO_L12N_T1_MRCC_34 U19 IO34_L12_N
IO_L13P_T2_MRCC_34 N18 IO34_L13_P
IO_L13N_T2_MRCC_34 P19 IO34_L13_N
IO_L14P_T2_SRCC_34 N20 IO34_L14_P
IO_L14N_T2_SRCC_34 P20 IO34_L14_N
IO_L15P_T2_DQS_34 T20 IO34_L15_P
IO_L15N_T2_DQS_34 U20 IO34_L15_N
IO_L16P_T2_34 V20 IO34_L16_P
IO_L16N_T2_34 W20 IO34_L16_N
IO_L17P_T2_34 Y18 IO34_L17_P
IO_L17N_T2_34 Y19 IO34_L17_N
IO_L18P_T2_34 V16 IO34_L18_P
IO_L18N_T2_34 W16 IO34_L18_N
IO_L19P_T3_34 R16 IO34_L19_P
IO_L19N_T3_VREF_34 R17 IO34_L19_N
IO_L20P_T3_34 T17 IO34_L20_P
IO_L20N_T3_34 R18 IO34_L20_N
IO_L21P_T3_DQS_34 V17 IO34_L21_P
IO_L21N_T3_DQS_34 V18 IO34_L21_N
IO_L22P_T3_34 W18 IO34_L22_P
IO_L22N_T3_34 W19 IO34_L22_N
IO_L23P_T3_34 N17 IO34_L23_P
IO_L23N_T3_34 P18 IO34_L23_N
IO_L24P_T3_34 P15 IO34_L24_P
IO_L24N_T3_34 P16 IO34_L24_N
IO_25_34 T19
XC7Z020-2CLG400I
U1C
BANK 35
IO_0_35 G14
IO_L1P_T0_AD0P_35 C20 IO35_L1_P
IO_L1N_T0_AD0N_35 B20 IO35_L1_N
IO_L2P_T0_AD8P_35 B19 IO35_L2_P
IO_L2N_T0_AD8N_35 A20 IO35_L2_N
IO_L3P_T0_DQS_AD1P_35 E17 IO35_L3_P
IO_L3N_T0_DQS_AD1N_35 D18 IO35_L3_N
IO_L4P_T0_35 D19 IO35_L4_P
IO_L4N_T0_35 D20 IO35_L4_N
IO_L5P_T0_AD9P_35 E18 IO35_L5_P
IO_L5N_T0_AD9N_35 E19 IO35_L5_N
IO_L6P_T0_35 F16 IO35_L6_P
IO_L6N_T0_VREF_35 F17 IO35_L6_N
IO_L7P_T1_AD2P_35 M19 IO35_L7_P
IO_L7N_T1_AD2N_35 M20 IO35_L7_N
IO_L8P_T1_AD10P_35 M17 IO35_L8_P
IO_L8N_T1_AD10N_35 M18 IO35_L8_N
IO_L9P_T1_DQS_AD3P_35 L19 IO35_L9_P
IO_L9N_T1_DQS_AD3N_35 L20 IO35_L9_N
IO_L10P_T1_AD11P_35 K19 IO35_L10_P
IO_L10N_T1_AD11N_35 J19 IO35_L10_N
IO_L11P_T1_SRCC_35 L16 IO35_L11_P
IO_L11N_T1_SRCC_35 L17 IO35_L11_N
IO_L12P_T1_MRCC_35 K17 IO35_L12_P
IO_L12N_T1_MRCC_35 K18 IO35_L12_N
IO_L13P_T2_MRCC_35 H16 IO35_L13_P
IO_L13N_T2_MRCC_35 H17 IO35_L13_N
IO_L14P_T2_AD4P_SRCC_35 J18 IO35_L14_P
IO_L14N_T2_AD4N_SRCC_35 H18 IO35_L14_N
IO_L15P_T2_DQS_AD12P_35 F19 IO35_L15_P
IO_L15N_T2_DQS_AD12N_35 F20 IO35_L15_N
IO_L16P_T2_35 G17 IO35_L16_P
IO_L16N_T2_35 G18 IO35_L16_N
IO_L17P_T2_AD5P_35 J20 IO35_L17_P
IO_L17N_T2_AD5N_35 H20 IO35_L17_N
IO_L18P_T2_AD13P_35 G19 IO35_L18_P
IO_L18N_T2_AD13N_35 G20 IO35_L18_N
IO_L19P_T3_35 H15 IO35_L19_P
IO_L19N_T3_VREF_35 G15 IO35_L19_N
IO_L20P_T3_AD6P_35 K14 IO35_L20_P
IO_L20N_T3_AD6N_35 J14 IO35_L20_N
IO_L21P_T3_DQS_AD14P_35 N15 IO35_L21_P
IO_L21N_T3_DQS_AD14N_35 N16 IO35_L21_N
IO_L22P_T3_AD7P_35 L14 IO35_L22_P
IO_L22N_T3_AD7N_35 L15 IO35_L22_N
IO_L23P_T3_35 M14 IO35_L23_P
IO_L23N_T3_35 M15 IO35_L23_N
IO_L24P_T3_AD15P_35 K16 IO35_L24_P
IO_L24N_T3_AD15N_35 J16 IO35_L24_N
IO_25_35 J15
XC7Z020-2CLG400I
U1A
BANK 13
IO_L6N_T0_VREF_13 V5
IO_L11P_T1_SRCC_13 U7 IO13_L11_P
IO_L11N_T1_SRCC_13 V7 IO13_L11_N
IO_L12P_T1_MRCC_13 T9 IO13_L12_P
IO_L12N_T1_MRCC_13 U10 IO13_L12_N
IO_L13P_T2_MRCC_13 Y7 IO13_L13_P
IO_L13N_T2_MRCC_13 Y6 IO13_L13_N
IO_L14P_T2_SRCC_13 Y9 IO13_L14_P
IO_L14N_T2_SRCC_13 Y8 IO13_L14_N
IO_L15P_T2_DQS_13 V8 IO13_L15_P
IO_L15N_T2_DQS_13 W8 IO13_L15_N
IO_L16P_T2_13 W10 IO13_L16_P
IO_L16N_T2_13 W9 IO13_L16_N
IO_L17P_T2_13 U9 IO13_L17_P
IO_L17N_T2_13 U8 IO13_L17_N
IO_L18P_T2_13 W11 IO13_L18_P
IO_L18N_T2_13 Y11 IO13_L18_N
IO_L19P_T3_13 T5 IO13_L19_P
IO_L19N_T3_VREF_13 U5 IO13_L19_N
IO_L20P_T3_13 Y12 IO13_L20_P
IO_L20N_T3_13 Y13 IO13_L20_N
IO_L21P_T3_DQS_13 V11 IO13_L21_P
IO_L21N_T3_DQS_13 V10 IO13_L21_N
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IO_L22N_T3_13 W6 IO13_L22_N
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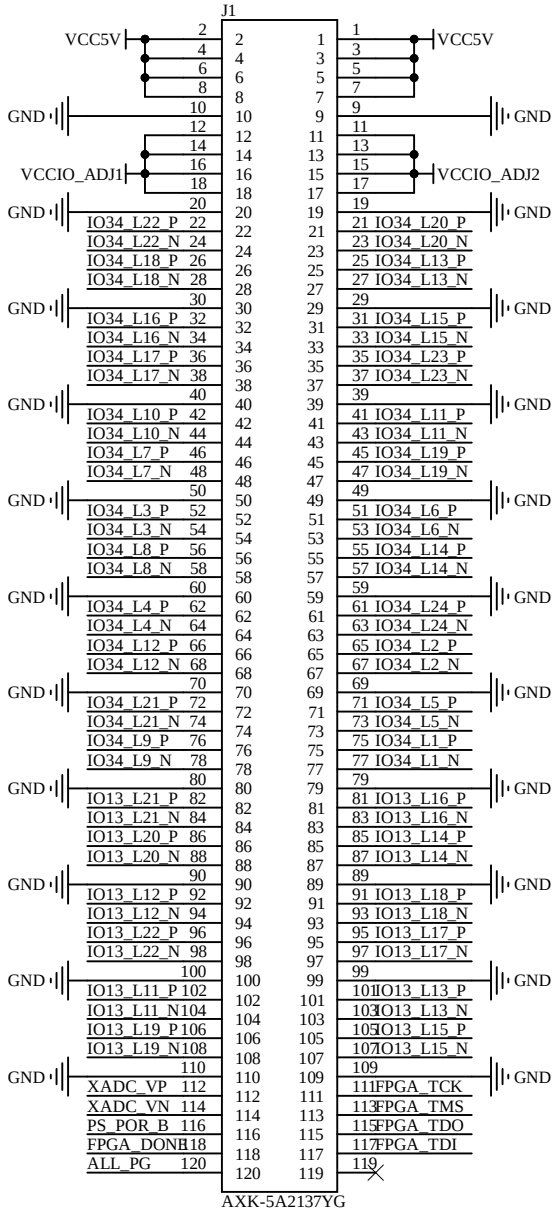


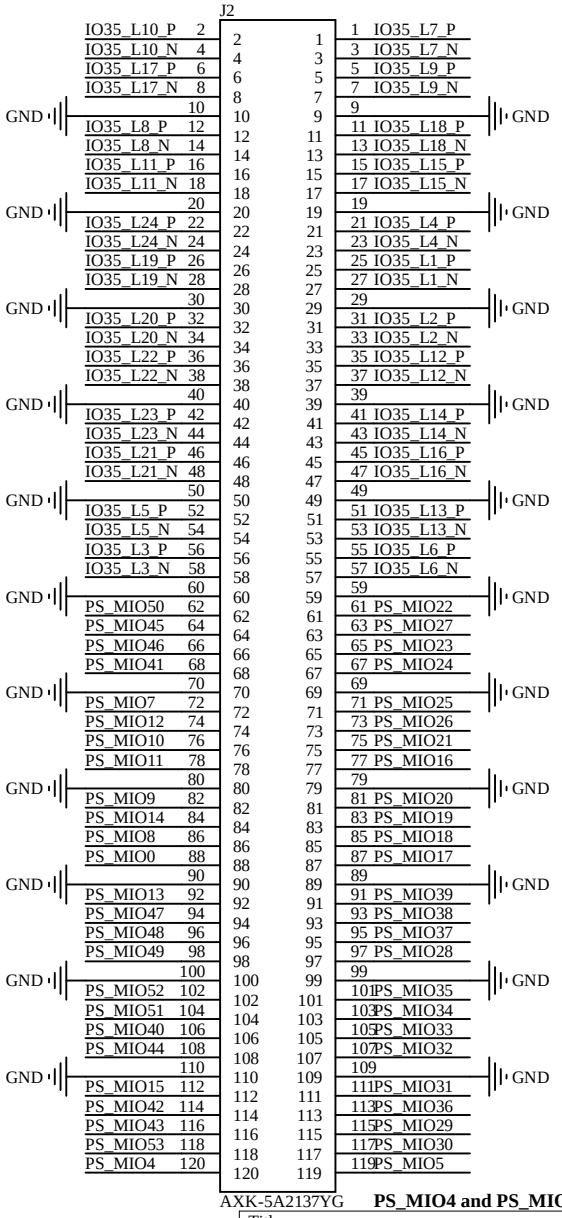
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DDR_VTT
R17 49.9R DDR3_A0
R18 49.9R DDR3_A1
R19 49.9R DDR3_A2
R20 49.9R DDR3_A3
R22 49.9R DDR3_A4
R23 49.9R DDR3_A5
R24 49.9R DDR3_A6
R25 49.9R DDR3_A7
R26 49.9R DDR3_A8
R27 49.9R DDR3_A9
R28 49.9R DDR3_A10
R29 49.9R DDR3_A11
R30 49.9R DDR3_A12
R31 49.9R DDR3_A13
R32 49.9R DDR3_A14
R34 49.9R DDR3_BA0
R37 49.9R DDR3_BA1
R38 49.9R DDR3_BA2
R39 49.9R DDR3_CKE
R40 49.9R DDR3_SO
R41 49.9R DDR3_RAS
R42 49.9R DDR3_CAS
R43 49.9R DDR3_WE
R44 49.9R DDR3_ODT
DDR3_CLK_P
R21 100R
DDR3_CLK_N
C45 10uF
C46 100nF
C47 100nF
C48 100nF
C49 10nF
GND
DDR_VTT
C50 10uF
C53 100nF
C56 100nF
C57 100nF
C58 10nF
VCC1V35
C16 4.7uF
C17 470nF
C18 470nF
C19 470nF
C20 470nF
C21 470nF
C22 470nF
C23 470nF
C24 100nF
C25 100nF
C26 100nF
C27 100nF
C28 100nF
C29 100nF
C30 4.7uF
C31 4.7uF
C32 4.7uF
C33 1uF
C34 1uF
C35 47nF
C36 47nF
C37 47nF
C38 47nF
C39 47nF
C40 47nF
C41 47nF
C42 47nF
C43 47nF
C44 47nF
GND
U3
DDR3_A0 DDR3_A1 DDR3_A2 DDR3_A3 DDR3_A4 DDR3_A5 DDR3_A6 DDR3_A7 DDR3_A8 DDR3_A9 DDR3_A10 DDR3_A11 DDR3_A12 DDR3_A13 DDR3_A14
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10/AP A11 A12/BC# A13 A14 NC
DDR3_BA0 DDR3_BA1 DDR3_BA2
BA0 BA1 BA2
DDR3_WE DDR3_RAS DDR3_CAS DDR3_CKE DDR3_SO
WE# RAS# CAS# CKE CS#
VREFDQ ODT VREFCA
RESET# ZQ
VDDQ VDD
VSS VSSQ
DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7 DQ8 DQ9 DQ10 DQ11 DQ12 DQ13 DQ14 DQ15
DDR3_D19 DDR3_D21 DDR3_D16 DDR3_D20 DDR3_D17 DDR3_D23 DDR3_D18 DDR3_D22 DDR3_D25 DDR3_D24 DDR3_D28 DDR3_D29 DDR3_D27 DDR3_D31 DDR3_D26 DDR3_D30
CK CK# LDQS LDQS# UDQS UDQS# LDM UDM
DDR3_CLK_P DDR3_CLK_N DDR3_DQS2_P DDR3_DQS2_N DDR3_DQS3_P DDR3_DQS3_N DDR3_DM2 DDR3_DM3
NC
MT41K256M16TW-107IT:P
VCC1V35
VTTREF
C52 10nF
C55 10nF
DDR3_ODT
R33 4.7K
R35 240R
DDR3_RESET
GND
U2
DDR3_A0 DDR3_A1 DDR3_A2 DDR3_A3 DDR3_A4 DDR3_A5 DDR3_A6 DDR3_A7 DDR3_A8 DDR3_A9 DDR3_A10 DDR3_A11 DDR3_A12 DDR3_A13 DDR3_A14
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10/AP A11 A12/BC# A13 A14 NC
DDR3_BA0 DDR3_BA1 DDR3_BA2
BA0 BA1 BA2
DDR3_WE DDR3_RAS DDR3_CAS DDR3_CKE DDR3_SO
WE# RAS# CAS# CKE CS#
VREFDQ ODT VREFCA
RESET# ZQ
VDDQ VDD
VSS VSSQ
DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7 DQ8 DQ9 DQ10 DQ11 DQ12 DQ13 DQ14 DQ15
DDR3_D2 DDR3_D0 DDR3_D1 DDR3_D4 DDR3_D6 DDR3_D7 DDR3_D3 DDR3_D5 DDR3_D12 DDR3_D11 DDR3_D15 DDR3_D9 DDR3_D14 DDR3_D10 DDR3_D13 DDR3_D8
CK CK# LDQS LDQS# UDQS UDQS# LDM UDM
DDR3_CLK_P DDR3_CLK_N DDR3_DQS0_P DDR3_DQS0_N DDR3_DQS1_P DDR3_DQS1_N DDR3_DM0 DDR3_DM1
NC
MT41K256M16TW-107IT:P
VCC1V35
VTTREF
C51 10nF
C54 10nF
DDR3_ODT
DDR3_RESET
R36 240R
GND



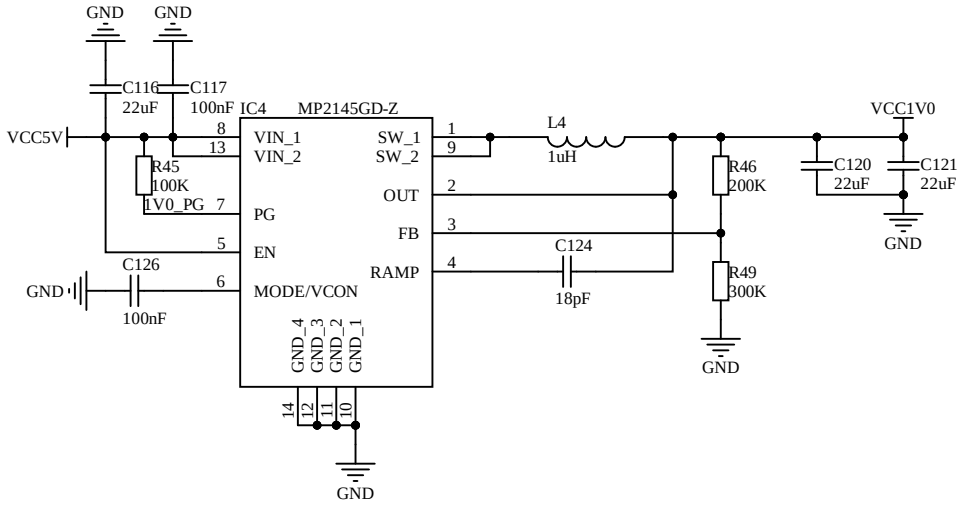
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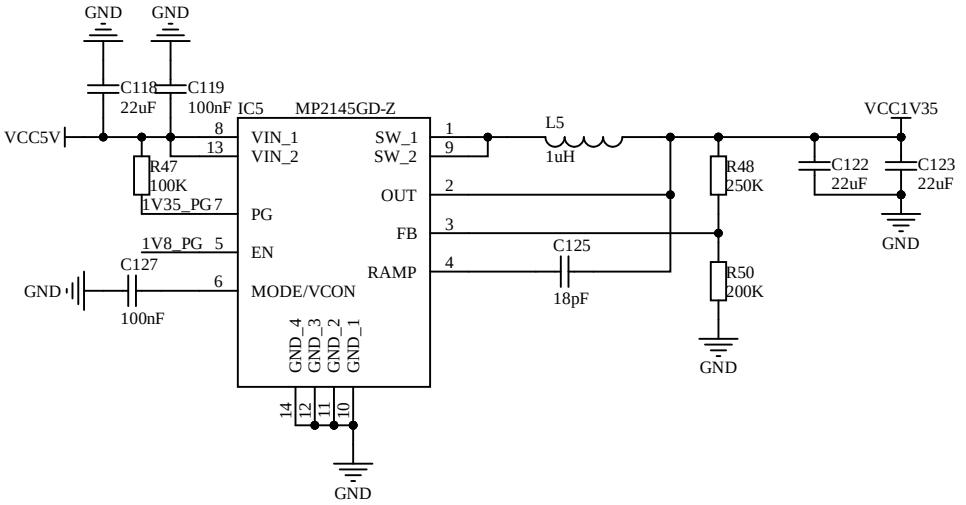


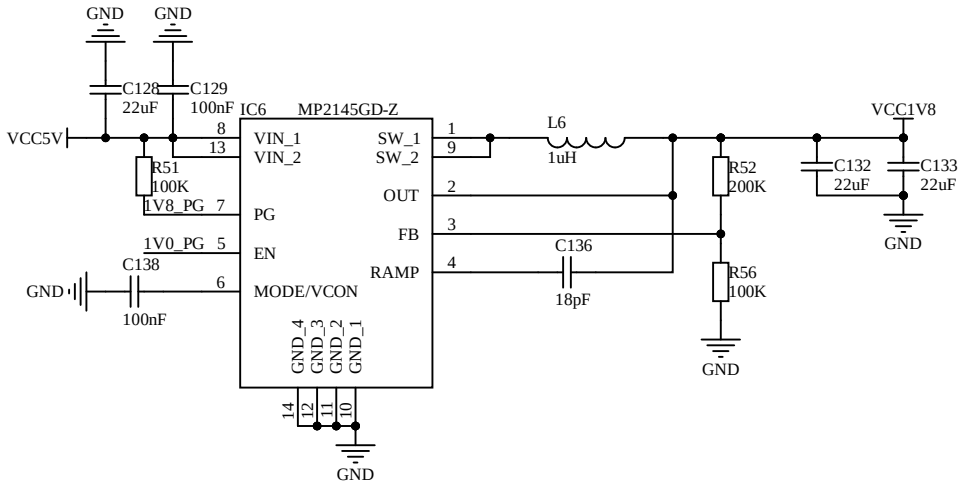


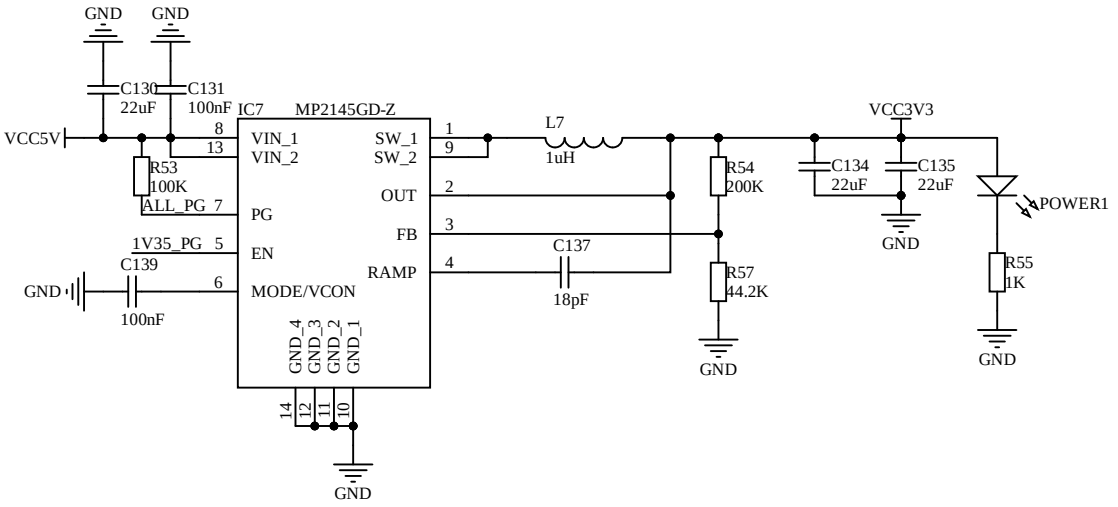
PS_MIO4 and PS_MIO5: no length matching

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